The Journal of Open Source Software

GraphCalc: A Python Package for Computing Graph Invariants in Automated Conjecturing Systems

Randy Davila [1,2*]

1 Department of Computational & Applied Mathematics, Rice University, United States **2** RelationalAI, United States * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 30 May 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

`GraphCalc` is a lightweight Python library for efficiently computing a wide range of graph invariants. It plays a central role in the **TxGraffiti** family of automated conjecturing systems, where it serves as the primary computational engine for generating data-driven mathematical conjectures in graph theory. The package provides accessible tools for calculating over 50 graph invariants, including many NP-hard invariants, such as the chromatic number, the independence number, and domination numbers.

Statement of Need

Modern research in graph theory and network science often requires the rapid computation of multiple graph invariants across large collections of graphs. While libraries such as NetworkX (Hagberg et al., 2008) and igraph provide graph traversal and manipulation utilities, they often lack built-in support for specialized invariants central to theoretical graph research — such as NP-hard invariants like the independence number or chromatic number.

`GraphCalc` fills this gap by offering a unified interface for computing classical and specialized invariants. It is **purpose-built for use in automated conjecturing frameworks**, particularly **TxGraffiti** (Davila, 2025a) and its variants (The `Optimist` (Davila, 2025b), etc.), which rely on precise invariant computations across thousands of graphs for conjecture generation and evaluation.

Features

`GraphCalc` supports both `networkx` graph objects and its own built-in graph types. This dual compatibility allows users to take advantage of familiar graph creation tools while benefiting from specialized features and optimizations specific to `GraphCalc`'s native graph representations.

- **Core invariants**: Computes over 50 classical and advanced graph invariants.
- **Graph filters**: Boolean functions to test properties like planarity, connectivity, and claw-freeness among many others.
- **Documentation and testing**: Full online documentation, type hints, and unit test coverage.

Example Usage

```
import graphcalc as gc
```

```
G = gc.petersen_graph()
gc.chromatic_number(G)       # Output: 3
gc.independence_number(G)      # Output: 4
gc.connected(G)           # Output: True
```

Relevance to Automated Discovery

GraphCalc is not merely a utility—it is an **enabling technology** in the development of AI systems that perform **automated mathematical reasoning**. In systems like TxGraffiti(Brimkov et al., 2024), conjectures are derived from numerical relationships among invariants computed by GraphCalc. The effectiveness of these conjecturing agents hinges directly on the reliability and breadth of invariant computations provided by this package.

Acknowledgements

The authors would like to acknowledge David Amos. Without his assistance this package would have not been possible.

References

Citations for tools and prior work referenced in this manuscript are included in the accompanying paper.bib file.

Brimkov, B., Davila, R., Schuerger, H., & Young, M. (2024). Computer assisted discovery: Zero forcing vs vertex cover. *Discrete Applied Mathematics*, *359*, 290–302.

Davila, R. (2025a). *Automated conjecturing in mathematics with TxGraffiti*.

Davila, R. (2025b). *The optimist: Towards fully automated graph theory research*.

Hagberg, A. A., Schult, D. A., & Swart, P. J. (2008). Exploring network structure, dynamics, and function using NetworkX. *Proceedings of the 7th Python in Science Conference (SciPy2008)*, 11–15.

DRAFT